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On the diophantine equation $x^2 - Dy^2 = n$

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Abstract We propose a method to determine the solvability of the diophantine equation $x^2 - Dy^2 = n$ for the following two cases:

(1) $D = pq$, where $p, q \equiv 1 \bmod 4$ are distinct primes with $(\frac{q}{p}) = 1$ and $(\frac{p}{q})_4(\frac{q}{p})_4 = -1$.

(2) $D = 2p_1p_2\cdots p_m$, where $p_i \equiv 1 \bmod 8, 1 \leqslant i \leqslant m$ are distinct primes and $D = r^2+s^2$ with $r, s \equiv \pm 3 \bmod 8$.

Keywords torus, Hilbert class field, reciprocity law

MSC(2010) 11D09, 11E12

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1 Introduction

Let D be a non-square integer. The question of whether the equation

$$x^2 - Dy^2 = n, \quad n \in \mathbb{Z}, \tag{1.1}$$

has an integral solution is a very old one (see [3]). We may recast the question in the language of algebraic geometry and ask whether the affine scheme over $\mathbb{Z}$ defined by (1.1) has an integral point. It is well known that the generic fiber of this affine scheme is a principal homogeneous space of tori when $n \neq 0$. Recently, Harari [6] showed that the Brauer-Manin obstruction is the only obstruction for the existence of integral points of such scheme. Xu and the author gave another proof of the result in [17, 18]. In this paper we consider the solvability of (1.1) by using the method in [17].

It should be pointed out that the method in [17] only produces the idelic class groups of $\mathbb{Q}(\sqrt{D})$. And these idelic class groups are not unique. In order to get the explicit conditions for the solvability, one needs further to construct the explicit abelian extensions of $\mathbb{Q}(\sqrt{D})$ corresponding to the idelic class groups. Such explicit construction is a wide open problem in general.

Notation and terminology are standard if not explained. Let $E = \mathbb{Q}(\sqrt{D})$ and $\mathfrak{o}_E$ be the ring of integers of E, Ω_E the set of places of E and ∞ the set of infinite places of E. Let $E_\mathfrak{p}$ be the completion of E at $\mathfrak{p}$ and $\mathfrak{o}_{E_\mathfrak{p}}$ the local completion of $\mathfrak{o}_E$ at $\mathfrak{p}$ for each $\mathfrak{p} \in \Omega_E$. Write $\mathfrak{o}_{E_\mathfrak{p}} = E_\mathfrak{p}$ for $\mathfrak{p} \in \infty$ and denote the adele ring (resp. the idele group) of E by $\mathbb{A}_E$ (resp. $\mathbb{I}_E$).

Let $\mathbf{X}_{(n)}$ denote the affine scheme over $\mathbb{Z}$ defined by $x^2 - Dy^2 = n$ for a non-zero integer n. Let $X_{(n)} = \mathbf{X}_{(n)} \times_\mathbb{Z} \mathbb{Q}$. Obviously $f = x + y\sqrt{D}$ is an invertible function on $X_{(n)} \otimes_\mathbb{Q} E$. And f induces a natural map

$$f_E : \ X_{(n)}(\mathbb{A}_\mathbb{Q}) \to \mathbb{I}_E.$$

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The restriction to $X_{(n)}(\mathbb{Q}_p)$ of f_E can be written by

$$f_E[(x_p,y_p)]=\begin{cases}(x_p+y_p\sqrt{D},x_p-y_p\sqrt{D}), & \text{if } p \text{ splits in } E/\mathbb{Q},\\ x_p+y_p\sqrt{D}, & \text{otherwise.}\end{cases}$$

Definition 1.1. Let $K_1,K_2,\ldots,K_m$ be finite abelian extensions over E. Let

$$\psi_{K_i/E}:\mathbb{I}_E\to \mathrm{Gal}(K_i/E),\quad \text{for } 1\leqslant i\leqslant m,$$

be the Artin map. We say that n satisfies the Artin condition of $K_1,K_2,\ldots,K_m$ if there is

$$\prod_{p\leqslant\infty}(x_p,y_p)\in\prod_{p\leqslant\infty}\mathbf{X}_{(n)}(\mathbb{Z}_p)$$

such that

$$\psi_{K_i/E}\Big(f_E\Big[\prod_{p\leqslant\infty}(x_p,y_p)\Big]\Big)=1_i,\quad \text{for } i=1,2,\ldots,m,$$

where 1_i is the identity element of $\mathrm{Gal}(K_i/E)$.

By the class field theory, it is a necessary condition for $\mathbf{X}_{(n)}(\mathbb{Z})\neq\emptyset$ that n satisfies the Artin condition of $K_1,K_2,\ldots,K_m$. And there is a finite abelian extension K/E that is independent on n, such that the Artin condition of K is also sufficient for $\mathbf{X}_{(n)}(\mathbb{Z})\neq\emptyset$ (see [17, Corollary 2.8]). For example, let $L=\mathbb{Z}+\mathbb{Z}\sqrt{D}$ and H_L be the ring class field corresponding to the order L, then the Artin condition of H_L is sufficient for $\mathbf{X}_{(n)}(\mathbb{Z})\neq\emptyset$ if $D<0$ (see [17, Proposition 3.1]). However, the Artin condition of H_L is not always sufficient for general D.

Let $E=\mathbb{Q}(\sqrt{D})$ and let T be the torus $R^1_{E/F}(\mathbb{G}_m)=\mathrm{Ker}[R_{E/F}(\mathbb{G}_{m,E})\to\mathbb{G}_{m,\mathbb{Q}}]$, where R denotes the Weil's restriction (see [8, p. 225]). Denote λ to be the embedding from T to $R_{E/F}(\mathbb{G}_{m,E})$. Obviously, λ induces a natural group homomorphism

$$\lambda_E:\ T(\mathbb{A}_F)\to\mathbb{I}_E.$$

Let $\mathbf{T}$ be the group scheme over $\mathbb{Z}$ defined by $x^2-Dy^2=1$ and let $T=\mathbf{T}\times_{\mathbb{Z}}\mathbb{Q}$. The generic fiber of $\mathbf{X}_{(n)}$ is a principal homogenous space of the torus T. Since $\mathbf{T}$ is separated over $\mathbb{Z}$, we can view $\mathbf{T}(\mathbb{Z}_p)$ as a subgroup of $T(\mathbb{Q}_p)$. The following result can be founded in [17, Proposition 1.4].

Proposition 1.2. *Let K_1/E and K_2/E be finite abelian extensions such that the group homomorphism induced by λ_E*

$$\bar{\lambda}_E:{}^{T(\mathbb{A}_{\mathbb{Q}})}\!/_{T(\mathbb{Q})\prod_{p\leqslant\infty}\mathbf{T}(\mathbb{Z}_p)}\longrightarrow {}^{\mathbb{I}_E}\!/_{E^*N_{K_1/E}(\mathbb{I}_{K_1})}\times{}^{\mathbb{I}_E}\!/_{E^*N_{K_2/E}(\mathbb{I}_{K_2})}$$

is well-defined and injective, where the well-defined means

$$\lambda_E\Big(T(\mathbb{Q})\prod_{p\leqslant\infty}\mathbf{T}(\mathbb{Z}_p)\Big)\subset(E^*N_{K_1/E}(\mathbb{I}_{K_1}))\cap(E^*N_{K_2/E}(\mathbb{I}_{K_2})).$$

Then $\mathbf{X}_{(n)}(\mathbb{Z})\neq\emptyset$ if and only if n satisfies the Artin condition of K_1 and K_2.

Let $D=pq$, here p and q are distinct primes. First we consider the solvability of $x^2-pqy^2=n, n\in\mathbb{Z}$. If $p\equiv 3 \mod 4$ or $q\equiv 3 \mod 4$, then $x^2-pqy^2=-1$ is not solvable over $\mathbb{Z}_2$. If p and q are of the form $4k+1$ and $(\frac{p}{q})=-1$, then $x^2-pqy^2=-1$ is solvable over $\mathbb{Z}$ (see [4, p. 228]). If p and q are distinct primes of the form $4k+1$ with $(\frac{p}{q})=1$ and $(\frac{p}{q})_4=(\frac{q}{p})_4=-1$, then $x^2-pqy^2=-1$ is also solvable over $\mathbb{Z}$ ([4], pp 228). For the above three cases, the equation $x^2-pqy^2=n$ is solvable over $\mathbb{Z}$ if and only if n satisfies the Artin condition of H_L by [17, Proposition 5.1]. In this paper we consider the case (in Section 1): $p,q\equiv 1 \mod 4$ are distinct primes, $(\frac{q}{p})=1$ and $(\frac{p}{q})_4(\frac{q}{p})_4=-1$.

The equation $x^2-pqy^2=pz^2$ is solvable over $\mathbb{Z}$ by Hasse principle. Fix an integral solution (x_0,y_0,z_0) of the equation such that $x_0>0$ and $(x_0,y_0)=1$. Let $\Theta=E(\sqrt{x_0-y_0\sqrt{pq}})$. We have the following result:

Theorem 1.3. *Let p and q be distinct primes of the form $4k+1$. Suppose $(\frac{q}{p})=1$ and $(\frac{p}{q})_4(\frac{q}{p})_4=-1$. Then the diophantine equation $x^2-pqy^2=n$ is solvable over $\mathbb{Z}$ if and only if n satisfies the Artin condition of H_L and Θ, where H_L is the ring class field corresponding to $L=\mathbb{Z}[\sqrt{pq}]$ and Θ is the quadratic extension of E defined as above.*

Let $D=2d$ and let $d=p_1p_2\cdots p_m$ where $p_i\equiv 1 \bmod 8, 1\leqslant i\leqslant m$, are distinct primes. The equation $x^2-2dy^2=2z^2$ is solvable over $\mathbb{Z}$ by the Hasse principle. Fix an integral solution (x_0,y_0,z_0) of the equation such that $x_0>0$ and $(x_0,y_0)=1$. Let $\Theta=E(\sqrt{x_0-y_0\sqrt{2d}})$.

Theorem 1.4. *Suppose that $p_i\equiv 1 \bmod 8, 1\leqslant i\leqslant m$ are distinct primes. Let $d=p_1p_2\cdots p_m$ such that $2d=r^2+s^2$ with $r,s\equiv\pm 3 \bmod 8$. Then the diophantine equation $x^2-2dy^2=n$ is solvable over $\mathbb{Z}$ if and only if n satisfies the Artin condition of H and Θ, where H is the Hilbert class field of E and Θ is the quadratic extension of E defined as above.*

As applications of Theorem 1.4, the solvability of the equations $x^2-dy^2=-1,\pm 2$ is considered in Section 4.

2 For the case $D=pq$

Let p and q be distinct primes of the form $4k+1$ with $(\frac{q}{p})=1$. Then the equation $x^2-pqy^2=pz^2$ is solvable over $\mathbb{Z}$ by the Hasse principle. Fix an integral solution (x_0,y_0,z_0) of the equation such that $x_0>0$ and $(x_0,y_0)=1$. Let $E=\mathbb{Q}(\sqrt{pq})$ and $\Theta=E(\sqrt{x_0-y_0\sqrt{pq}})$. Then Θ is totally real and Θ/E is unramified over all primes except the primes over $2p$.

Lemma 2.1. *Let p and q be distinct primes of the form $4k+1$ with $(\frac{q}{p})=1$. Let $l=2$ or p. If x_l and y_l in $\mathbb{Z}_l$ satisfy $x_l^2-pqy_l^2=1$, then the quadratic Hilbert symbol*

$$\prod_{v|l}\left(\frac{x_l-y_l\sqrt{pq},x_0-y_0\sqrt{pq}}{v}\right)=1,$$

where $v\in\Omega_E$.

Proof. (1) Assume $l=2$ is split in $E/\mathbb{Q}$. Then we have

$$\begin{aligned}\prod_{v|2}\left(\frac{x_2-y_2\sqrt{pq},x_0-y_0\sqrt{pq}}{v}\right)&=\left(\frac{x_2-y_2\sqrt{pq},x_0-y_0\sqrt{pq}}{2}\right)\cdot\left(\frac{x_2+y_2\sqrt{pq},x_0+y_0\sqrt{pq}}{2}\right)\\&=\left(\frac{x_2-y_2\sqrt{pq},x_0-y_0\sqrt{pq}}{2}\right)\cdot\left(\frac{(x_2-y_2\sqrt{pq})^{-1},x_0+y_0\sqrt{pq}}{2}\right)\\&=\left(\frac{x_2-y_2\sqrt{pq},x_0-y_0\sqrt{pq}}{2}\right)\cdot\left(\frac{x_2-y_2\sqrt{pq},x_0+y_0\sqrt{pq}}{2}\right)\\&=\left(\frac{x_2-y_2\sqrt{pq},p}{2}\right)=1,\end{aligned}$$

the last equation holds since $x_2-y_2\sqrt{pq}$ is an unit in $\mathbb{Z}_2$ and $p\equiv 1 \bmod 4$.

(2) Assume 2 is inert in $E/\mathbb{Q}$. Let $l=2$ or p. Let $\xi=x_l+y_l\sqrt{pq}$. Let v be the unique place of E over l. Since $N_{E_v/\mathbb{Q}_l}(\xi)=1$, there exists $\alpha\in E_v^*$ such that $\xi=\sigma(\alpha)\alpha^{-1}$ by Hilbert's theorem 90, where σ is the non-trivial element in $\mathrm{Gal}(E_v/\mathbb{Q}_l)$. Then

$$\begin{aligned}\left(\frac{\xi,x_0-y_0\sqrt{pq}}{v}\right)&=\left(\frac{\sigma(\alpha)\alpha^{-1},x_0-y_0\sqrt{pq}}{v}\right)\\&=\left(\frac{N_{E_v/\mathbb{Q}_l}(\alpha),x_0-y_0\sqrt{pq}}{v}\right)=\left(\frac{N_{E_v/\mathbb{Q}_l}(\alpha),p}{l}\right).\end{aligned}$$

Suppose $l=2$. Since $E_v/\mathbb{Q}_2$ is unramified and $p\equiv 1 \bmod 4$, one has

$$\left(\frac{\xi,x_0-y_0\sqrt{pq}}{v}\right)=\left(\frac{N_{E_v/\mathbb{Q}_2}(\alpha),p}{2}\right)=1.$$

Suppose $l=p$. Since $(\frac{q}{p})=1$ and $E_v=\mathbb{Q}_p(\sqrt{pq})$, one has

$$\left(\frac{\xi, x_0-y_0\sqrt{pq}}{v}\right)=\left(\frac{N_{E_v/\mathbb{Q}_p}(\alpha),p}{p}\right)=\left(\frac{N_{E_v/\mathbb{Q}_p}(\alpha),pq}{p}\right)=1. \qquad \square$$

We make use of the following interesting result, due to Burde [2, p. 183]:

Lemma 2.2. *If $p=a^2+b^2, q=c^2+d^2, a\equiv c\equiv 1, b\equiv d\equiv 0 \pmod 2), a,b,c,d>0$, p and q are primes, and $(\frac{q}{p})=1$, then*

$$\left(\frac{p}{q}\right)_4\left(\frac{q}{p}\right)_4=(-1)^{(p-1)/4}\left(\frac{ad-bc}{p}\right).$$

Lemma 2.3. *Let p and q be distinct primes of the form $4k+1$. Suppose $(\frac{q}{p})=1$ and $(\frac{p}{q})_4(\frac{q}{p})_4=-1$. Let $l=2$ or p and let x_l and y_l in $\mathbb{Z}_l$ satisfy $x_l^2-pqy_l^2=-1$. Then the quadratic Hilbert symbol*

$$\prod_{l|2p}\prod_{v|l}\left(\frac{x_l+y_l\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)=-1,$$

where $v\in\Omega_E$.

Proof. Let $p=a^2+b^2, q=c^2+d^2, a\equiv c\equiv 1, b\equiv d\equiv 0 \pmod 2), a,b,c,d>0$. And let $r=ad-bc, s=ac+bd$. Then $r^2+s^2=pq$ and s is odd. Obviously $(r/s)^2\equiv -1 \mod p$. So

$$\left(\frac{s}{p}\right)=\left(\frac{r}{p}\right)\left(\frac{-1}{p}\right)_4=\left(\frac{r}{p}\right)\left(\frac{2}{p}\right)=\left(\frac{r}{p}\right)(-1)^{(p-1)/4}=\left(\frac{p}{q}\right)_4\left(\frac{q}{p}\right)_4=-1$$

by Lemma 2.2.

We have

$$\prod_{l|2p}\prod_{v|l}\left(\frac{x_l+y_l\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)=\prod_{v|2p}\left(\frac{(r-\sqrt{pq})s^{-1}, x_0-y_0\sqrt{pq}}{v}\right)$$

which uses Lemma 2.1. Therefore we have

$$\begin{aligned}\prod_{l|2p}\prod_{v|l}\left(\frac{x_l+y_l\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)&=\prod_{v|2p}\left(\frac{r-\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)\cdot\prod_{v|2p}\left(\frac{s, x_0-y_0\sqrt{pq}}{v}\right)\\&=\prod_{v|2p}\left(\frac{r-\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)\cdot\left(\frac{s, pz_0^2}{2}\right)\cdot\left(\frac{s, pz_0^2}{p}\right)\\&=-\prod_{v|2p}\left(\frac{r-\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right).\end{aligned}$$

By Hilbert reciprocity law, one has

$$\prod_{v|2p}\left(\frac{r-\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)=\prod_{\mathfrak{p}\nmid 2p}\left(\frac{r-\sqrt{pq}, x_0-y_0\sqrt{pq}}{\mathfrak{p}}\right).$$

Since

$$(r-\sqrt{pq})(r+\sqrt{pq})=-s^2 \quad\text{and}\quad (x_0-y_0\sqrt{pq})(x_0+y_0\sqrt{pq})=pz_0^2$$

with $(r,s)=1$ and $(x_0,y_0)=1$ respectively, one has

$$\mathrm{ord}_{\mathfrak{p}}(r-\sqrt{pq})\equiv\mathrm{ord}_{\mathfrak{p}}(x_0-y_0\sqrt{pq})\equiv 0 \mod 2$$

for $\mathfrak{p}\nmid 2p$ and $\mathfrak{p}<\infty_E$. Since $x_0-y_0\sqrt{pq}>0$ over $\mathfrak{p}\in\infty_E$, one obtains that

$$\left(\frac{r-\sqrt{pq}, x_0-y_0\sqrt{pq}}{\mathfrak{p}}\right)=1$$

for all $\mathfrak{p} \nmid 2p$. One concludes that

$$\prod_{l|2p}\prod_{v|l}\left(\frac{x_l+y_l\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)=-1. \qquad \square$$

The following result was first proved by Scholz [13] and was proved by Brown [1].

Corollary 2.4. *Let p and q be distinct primes of the form $4k+1$. Suppose $(\frac{q}{p})=1$ and $(\frac{p}{q})_4(\frac{q}{p})_4=-1$. Then the equation $x^2-pqy^2=-1$ is not solvable over $\mathbb{Z}$; the equation $x^2-pqy^2=p$ is solvable over $\mathbb{Z}$ if and only if $(\frac{q}{p})_4=1$.*

Proof. By Lemma 2.3, one has

$$\prod_{l|2p}\prod_{v|l}\left(\frac{x_l+y_l\sqrt{pq}, x_0-y_0\sqrt{pq}}{v}\right)=-1$$

for $(x_l, y_l)\in\mathbb{Z}_l\times\mathbb{Z}_l$ with $x_l^2-pqy_l^2=-1$. This implies that

$$\psi_{\Theta/E}\left(f_E\Big[\prod_{p\leqslant\infty}(x_p,y_p)\Big]\right)=-1$$

for any $\prod_{l\leqslant\infty}(x_l,y_l)\in\prod_{l\leqslant\infty}\mathbf{X}_{(-1)}(\mathbb{Z}_l)$, since Θ/E is unramified over each place v except $v\mid 2p$. Then $x^2-pqy^2=-1$ is not solvable over $\mathbb{Z}$ by the class field theory.

One and only one of the three equations

$$x^2-pqy^2=-1,\quad x^2-pqy^2=p,\quad x^2-pqy^2=q$$

is solvable over $\mathbb{Z}$ (see [4, p. 228]). And if $x^2-pqy^2=p$ is solvable over $\mathbb{Z}$, then $(\frac{q}{p})_4=1$ (see [4, p. 230]). If $(\frac{q}{p})_4=1$, then $(\frac{p}{q})_4=-1$. So $x^2-pqy^2=q$ is not solvable over $\mathbb{Z}$. And $x^2-pqy^2=-1$ is not solvable over $\mathbb{Z}$. Therefore $x^2-pqy^2=p$ is solvable over $\mathbb{Z}$. $\square$

By Lemmas 2.1 and 2.3, now we can give the proof of Theorem 1.3.

Proof. Let $L=\mathbb{Z}+\mathbb{Z}\sqrt{pq}$ be a ring. For any prime l, $E_l=E\otimes_{\mathbb{Q}}\mathbb{Q}_l$, L_l is the l-adic completion of L inside E_l and $L_l^\times$ is the unit group of L_l. Recall $T=R^1_{E/F}(\mathbb{G}_{m,E})$ and $\mathbf{T}$ is the group scheme over $\mathbb{Z}$ defined by $x^2-pqy^2=1$, we have

$$T(\mathbb{Q})=\{\xi\in E^*:\ N_{E/\mathbb{Q}}(\xi)=1\}$$

and

$$\mathbf{T}(\mathbb{Z}_l)=\{\xi\in L_l^\times:\ N_{E_l/\mathbb{Q}_l}(\xi)=1\}.$$

And $L_\infty^\times=E_\infty^*=\mathbb{R}^*\times\mathbb{R}^*$.

Let $l=2$ or p. By Lemma 2.1, one has

$$\prod_{v|l}\left(\frac{\xi, x_0-y_0\sqrt{pq}}{v}\right)=1,$$

where $\xi=x_l+y_l\sqrt{pq}$ and $(x_l,y_l)\in\mathbb{Z}_l\times\mathbb{Z}_l$ with $x_l^2-pqy_l^2=1$, where $v\in\Omega_E$. This implies that

$$\lambda_E(\mathbf{T}(\mathbb{Z}_l))\subseteq E^*N_{\Theta/E}(\mathbb{I}_\Theta).$$

Since Θ/E is unramified over each place v except $v\mid 2p$, the natural group homomorphism

$$\bar{\lambda}_E: {}^{T(\mathbb{A}_\mathbb{Q})}\big/_{T(\mathbb{Q})\prod_{l\leqslant\infty}\mathbf{T}(\mathbb{Z}_l)}\longrightarrow\big[{}^{\mathbb{I}_E}\big/_{E^*N_{\Theta/E}(\mathbb{I}_\Theta)}\big]\times\big[{}^{\mathbb{I}_E}\big/_{E^*\prod_{l\leqslant\infty}L_l^\times}\big]$$

is well-defined. By Proposition 1.2, we only need to show $\bar{\lambda}_E$ is injective.

Let $u \in \mathrm{Ker}\lambda_E$. Then there are $\alpha \in E^*$ and $i \in \prod_{l\leqslant\infty} L_l^\times$ with $\lambda_E(u) = \alpha i$. We have

$$N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i)^{-1} \in \mathbb{Q}^* \cap \left(\prod_{l\leqslant\infty} \mathbb{Z}_l^\times\right) = \{\pm 1\}.$$

If $N_{E/\mathbb{Q}}(\alpha) \neq 1$, one obtains $N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i) = -1$. Write $i = (i_v)_v \in \mathbb{I}_E$. Since Θ/E is unramified over each place v except $v \mid 2p$, one concludes that $\psi_{\Theta/E}(i_v)$ is trivial for all primes $v \nmid 2p$, where i_v is regarded as an idele whose v-component is i_v and 1 otherwise. One gets

$$\psi_{\Theta/E}(\alpha i) = \psi_{\Theta/E}(i) = \prod_{v|2p} \psi_{\Theta/E}(i_v) = -1$$

by Lemma 2.3, where $\psi_{\Theta/E} : \mathbb{I}_E \to \mathrm{Gal}(\Theta/E)$ is the Artin map. This contradicts to $u \in \mathrm{Ker}\lambda_E$.

Therefore $N_{E/\mathbb{Q}}(\alpha) = 1$, one concludes that

$$N_{E/\mathbb{Q}}(\alpha) = N_{E/\mathbb{Q}}(i) = 1 \Rightarrow \alpha \in T(\mathbb{Q}) \quad \text{and} \quad i \in \prod_{l\leqslant\infty} \mathbf{T}(\mathbb{Z}_l).$$

So $\alpha i \in T(\mathbb{Q}) \prod_{l\leqslant\infty} \mathbf{T}(\mathbb{Z}_l)$. □

Finally we will use Theorem 1.3 to give an explicit example. For any integer n, one can write $n = (-1)^{s_0} 2^{s_1} 13^{s_2} 17^{s_3} {p_1}^{e_1} \cdots {p_g}^{e_g}$ and $P(n) = \{p_1, \ldots, p_g\}$. Denote

$$P_1 = \left\{p \in P(n) : \left(\frac{13}{p}\right) = \left(\frac{17}{p}\right) = -1\right\}, \quad P_2 = \left\{p \in P(n) : \left(\frac{221}{p}\right) = -1\right\},$$
$$P_3 = \left\{p \in P(n) : \left(\frac{13}{p}\right) = \left(\frac{17}{p}\right) = 1 \text{ and } x^4 - 238x^2 + 17 \equiv 0 \bmod p \text{ is solvable}\right\}.$$

Let

$$n_1 = \prod_{p_i \in P(n)\setminus P_2} p_i^{e_i}.$$

Example 2.5. *Let n be an integer with the above notation. Then the equation*

$$x^2 - 221y^2 = n$$

is solvable over $\mathbb{Z}$ if and only if

(1) *s_1 is even, $(\frac{n_1}{17}) = 1$ and $(\frac{221}{p_i}) = 1$ for odd e_i.*

(2) *$P_1 \neq \emptyset$; or*

$$\prod_{p_i \in P_3} (-1)^{e_i} \prod_{p_i \in P(n)\setminus P_2} \left(\frac{-1}{p_i}\right)^{e_i} = (-1)^{s_0+s_2} \cdot \left(\frac{n_1}{17}\right)_4$$

for $P_1 = \emptyset$.

Proof. Since $(\frac{17}{13})_4 = -1$ and $(\frac{13}{17})_4 = 1$, Theorem 1.3 can be applied. The ring class field H_L associated to the subring $\mathbb{Z}[\sqrt{221}]$ is $\mathbb{Q}(\sqrt{13}, \sqrt{17})$. Since the equation $x^2 - 221y^2 = 17$ has an integral solution for $x = 119$ and $y = 8$, one can choose $\Theta = E(\sqrt{119 - 8\sqrt{221}})$. We can get this result by some computations. □

3 The case $D = 2d$

Let $d = p_1 \cdots p_m$, where $p_i \equiv \pm 1 \bmod 8, 1 \leqslant i \leqslant m$ are distinct primes and $d \equiv 1 \bmod 8$. The equation $x^2 - 2dy^2 = 2z^2$ is solvable over $\mathbb{Z}$ by Hasse principle. Fix an integral solution (x_0, y_0, z_0) of the equation such that $x_0 > 0$ and $(x_0, y_0) = 1$. Let $\Theta = E(\sqrt{x_0 - y_0\sqrt{2d}})$. Then Θ is totally real and Θ/E is unramified over all primes except the unique place over 2 and 2 is totally ramified in $\Theta/\mathbb{Q}$. In this section, we will prove Theorem 1.4. First the following lemmas will be proved.

Lemma 3.1. *Let $d = p_1p_2\cdots p_m$, where $p_i \equiv \pm 1 \mod 8, 1 \leqslant i \leqslant m$ are distinct primes and $d \equiv 1 \mod 8$. Then the quadratic Hilbert symbol*

$$\left(\frac{\xi, x_0 - y_0\sqrt{2d}}{v}\right) = 1$$

for any $\xi \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi) = 1$, where v is the unique place of E over 2 and (x_0, y_0) is given as above.

Proof. Since $N_{E_v/\mathbb{Q}_2}(\xi) = 1$, there exists $\alpha \in E_v^*$ such that $\xi = \sigma(\alpha)\alpha^{-1}$ by Hilbert's theorem 90, where σ is the non-trivial element in $\mathrm{Gal}(E_v/\mathbb{Q}_2)$. Then

$$\begin{aligned}\left(\frac{\xi, x_0 - y_0\sqrt{2d}}{v}\right) &= \left(\frac{\sigma(\alpha)\alpha^{-1}, x_0 - y_0\sqrt{2d}}{v}\right)\\ &= \left(\frac{N_{E_v/\mathbb{Q}_2}(\alpha), x_0 - y_0\sqrt{2d}}{v}\right) = \left(\frac{N_{E_v/\mathbb{Q}_2}(\alpha), 2}{2}\right).\end{aligned}$$

Since $d \equiv 1 \mod 8$ and $E_v = \mathbb{Q}_2(\sqrt{2d})$, one has

$$\left(\frac{\xi, x_0 - y_0\sqrt{2d}}{v}\right) = \left(\frac{N_{E_v/\mathbb{Q}_2}(\alpha), 2}{2}\right) = \left(\frac{N_{E_v/\mathbb{Q}_2}(\alpha), 2d}{2}\right) = 1. \qquad \square$$

Lemma 3.2. *Suppose that $p_i \equiv 1 \mod 8, 1 \leqslant i \leqslant m$ are distinct primes. Let $d = p_1p_2\cdots p_m$ such that $2d = r^2 + s^2$ with $r, s \equiv \pm 3 \mod 8$. If x_2 and y_2 in $\mathbb{Q}_2$ satisfy $x_2^2 - 2dy_2^2 = -1$, then the quadratic Hilbert symbol*

$$\left(\frac{x_2 + y_2\sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = -1,$$

where v is the unique place of E over 2 and (x_0, y_0) is given as above.

Proof. We have

$$\left(\frac{x_2 + y_2\sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = \left(\frac{(r - \sqrt{2d})s^{-1}, x_0 - y_0\sqrt{2d}}{v}\right),$$

which uses Lemma 3.1. Therefore

$$\begin{aligned}\left(\frac{x_2 + y_2\sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) &= \left(\frac{r - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right)\cdot\left(\frac{s, x_0 - y_0\sqrt{2d}}{v}\right)\\ &= \left(\frac{r - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right)\cdot\left(\frac{s, 2z_0^2}{2}\right)\\ &= -\left(\frac{r - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right).\end{aligned}$$

By Hilbert reciprocity law, one has

$$\left(\frac{r - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = \prod_{\mathfrak{p}\neq v}\left(\frac{r - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{\mathfrak{p}}\right).$$

Since

$$(r - \sqrt{2d})(r + \sqrt{2d}) = -s^2 \quad \text{and} \quad (x_0 - y_0\sqrt{2d})(x_0 + y_0\sqrt{2d}) = 2z_0^2$$

with $(r, s) = 1$ and $(x_0, y_0) = 1$ respectively, one has

$$\mathrm{ord}_{\mathfrak{p}}(r - \sqrt{2d}) \equiv \mathrm{ord}_{\mathfrak{p}}(x_0 - y_0\sqrt{2d}) \equiv 0 \mod 2$$

for $\mathfrak{p} \neq v$ and $\mathfrak{p} < \infty_E$. Since $x_0 - y_0\sqrt{2d} > 0$ over $\mathfrak{p} \in \infty_E$, one obtains that

$$\left(\frac{r - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{\mathfrak{p}}\right) = 1$$

for all $\mathfrak{p} \neq v$. One concludes that

$$\left(\frac{x_2 + y_2\sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = -1. \qquad \square$$

With a similar argument as in the proof of Theorem 1.3, we can prove Theorem 1.4 by Lemmas 3.1 and 3.2.

4 Some applications

In this section we consider the solvability of the equations $x^2 - 2dy^2 = -1, \pm 2$ by using Theorem 1.4. It's well known that at most one of the three equations

$$x^2 - 2dy^2 = -1, \quad x^2 - 2dy^2 = \pm 2$$

is solvable over $\mathbb{Z}$ (see [11, pp. 106–109]).

Lemma 4.1. *Suppose that $p_i \equiv \pm 1 \mod 8, 1 \leqslant i \leqslant m$ are distinct primes. Let $d = p_1 p_2 \cdots p_m \equiv 1 \mod 8$. If x_2 and y_2 in $\mathbb{Q}_2$ satisfy $x_2^2 - 2dy_2^2 = 2$, then the quadratic Hilbert symbol*

$$\left(\frac{x_2 + y_2\sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = \begin{cases} 1, & \text{if } d \equiv 1 \mod 16, \\ -1, & \text{if } d \equiv 9 \mod 16, \end{cases}$$

where v is the unique place of E over 2 and (x_0, y_0, z_0) is given as in Section 2, i.e., the integers x_0, y_0 and z_0 are relatively prime with $x_0 > 0$ and satisfy $x_0^2 - 2dy_0^2 = 2z_0^2$.

Proof. The equation $x^2 - 2y^2 = 2d$ is solvable over $\mathbb{Z}$ since it satisfies Hasse principle. Choose one solution (a, b) of the equation and obviously a is even and b is odd. Let $a = 2a'$. First we assume $d \equiv 1 \mod 16$. We will show $b \equiv \pm 1 \mod 8$. Otherwise we assume $b \equiv \pm 3 \mod 8$, then one has $a'^2 = (d + b^2)/2 \equiv 5 \mod 8$. It is impossible since $a'^2 \equiv 0, 1$ or $4 \mod 8$. Similarly we can prove $b \equiv \pm 3 \mod 8$ if $d \equiv 9 \mod 16$. Then we have

$$b \equiv \begin{cases} \pm 1 \mod 8, & \text{if } d \equiv 1 \mod 16, \\ \pm 3 \mod 8, & \text{if } d \equiv 9 \mod 16. \end{cases}$$

Let v be the unique place of E over 2. And let $\xi \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi) = 2$. By Lemma 3.1, we have

$$\begin{aligned} \left(\frac{\xi, x_0 - y_0\sqrt{2d}}{v}\right) &= \left(\frac{(a - \sqrt{2d})b^{-1}, x_0 - y_0\sqrt{2d}}{v}\right) \\ &= \left(\frac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) \cdot \left(\frac{b, x_0 - y_0\sqrt{2d}}{v}\right) \\ &= \left(\frac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) \cdot \left(\frac{b, 2z_0^2}{2}\right) \\ &= \begin{cases} \left(\dfrac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right), & \text{if } d \equiv 1 \mod 16, \\ -\left(\dfrac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right), & \text{if } d \equiv 9 \mod 16. \end{cases} \end{aligned}$$

Then we only need to show

$$\left(\frac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = 1.$$

By Hilbert reciprocity law, one has

$$\left(\frac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{v}\right) = \prod_{\mathfrak{p} \neq v} \left(\frac{a - \sqrt{2d}, x_0 - y_0\sqrt{2d}}{\mathfrak{p}}\right).$$

Since

$$(a-\sqrt{2d})(a+\sqrt{2d})=-2b^2 \quad \text{and} \quad (x_0-y_0\sqrt{2d})(x_0+y_0\sqrt{2d})=2z_0^2$$

with $(a,b)=1$ and $(x_0,y_0,z_0)=1$ respectively, one has

$$\mathrm{ord}_{\mathfrak{p}}(a-\sqrt{2d})\equiv \mathrm{ord}_{\mathfrak{p}}(x_0-y_0\sqrt{2d})\equiv 0 \bmod 2$$

for $\mathfrak{p}\neq v$ and $\mathfrak{p}<\infty_E$. Since $x_0-y_0\sqrt{2d}>0$ over $\mathfrak{p}\in\infty_E$, one obtains that

$$\left(\frac{a-\sqrt{2d},x_0-y_0\sqrt{2d}}{\mathfrak{p}}\right)=1$$

for all $\mathfrak{p}\neq v$. One concludes that

$$\left(\frac{a-\sqrt{2d},x_0-y_0\sqrt{2d}}{v}\right)=1. \qquad \square$$

Corollary 4.2. *Let d be a positive integer and $d\equiv 9 \bmod 16$, then $x^2-2dy^2=2$ is not solvable over $\mathbb{Z}$.*

Proof. Let $d=p_1^{e_1}\cdots p_m^{e_m}$ where $e_i, 1\leqslant i\leqslant m$ are positive integers and $p_i, 1\leqslant i\leqslant m$ are district primes. If there is some p_i satisfying $p_i\equiv\pm 3 \bmod 8$, the equation $x^2-2dy^2=2$ is not solvable over $\mathbb{Z}$ since $(\frac{2}{p_i})=-1$. Therefore we can assume $p_i\equiv\pm 1 \bmod 8$ for all i.

First we assume that d ($\equiv 9 \bmod 16$) is square-free. By Lemma 4.1, one has

$$\psi_{\Theta/E}\left(f_E\left[\prod_{p\leqslant\infty}(x_p,y_p)\right]\right)=\left(\frac{x_2+y_2\sqrt{2d},x_0-y_0\sqrt{2d}}{v}\right)=-1$$

for any $\prod_{p\leqslant\infty}(x_p,y_p)\in\prod_{p\leqslant\infty}\mathbf{X}_{(2)}(\mathbb{Z}_p)$, since $\Theta=E(\sqrt{x_0-y_0\sqrt{2d}})$ is unramifed over E everywhere except v. Then $x^2-2dy^2=2$ is not solvable over $\mathbb{Z}$ by Theorem 1.4.

For general $d=p_1^{e_1}\cdots p_m^{e_m}$ with $p_i\equiv\pm 1 \bmod 8$ and $e_i\geqslant 1$ for all i, we can write $d=d'\cdot m^2$ with d' is square-free and $m\equiv\pm 1 \bmod 8$. Then we can see $d'\equiv d\equiv 9 \bmod 16$. Assume $x^2-dy^2=2$ is solvable over $\mathbb{Z}$. Then $x^2-d'y^2=2$ is solvable over $\mathbb{Z}$, which is contrary to the above arguments. Therefore we have that $x^2-dy^2=2$ is not solvable over $\mathbb{Z}$. $\square$

Remark. For the special case that $d\equiv 9 \bmod 16$ is a prime, the unsolvability of $x^2-2dy^2=2$ was proved by Pall (see [10]).

Lemma 4.3. *Suppose that $p_i\equiv 1 \bmod 8, 1\leqslant i\leqslant m$ are distinct primes. Let $d=p_1p_2\cdots p_m$. If x_2 and y_2 in $\mathbb{Q}_2$ satisfy $x_2^2-2dy_2^2=-2$, then the quadratic Hilbert symbol*

$$\left(\frac{x_2+y_2\sqrt{2d},x_0-y_0\sqrt{2d}}{v}\right)=\left(\frac{2}{d}\right)_4,$$

where v is the unique place of E over 2 and (x_0,y_0) is given as in Section 2.

Proof. Since $d=p_1\cdots p_m$ and $p_i\equiv 1 \bmod 8, 1\leqslant i\leqslant m$, we have $x^2+2y^2=2d$ is solvable over $\mathbb{Z}$ since it satisfies Hasse principle. Choose one solution (a,b) of the equation and let $\eta=(a-\sqrt{2d})/b$. Then $N_{E_v/\mathbb{Q}_2}(\eta)=-2$. By Lemma 3.1 we have

$$\left(\frac{x_2+y_2\sqrt{2d},x_0-y_0\sqrt{2d}}{v}\right)=\left(\frac{\eta,x_0-y_0\sqrt{2d}}{v}\right).$$

So we only need to show

$$\left(\frac{\eta,x_0-y_0\sqrt{2d}}{v}\right)=\left(\frac{2}{d}\right)_4.$$

The Hilbert symbol

$$\left(\frac{\eta,x_0-y_0\sqrt{2d}}{v}\right)=\left(\frac{a-\sqrt{2d},x_0-y_0\sqrt{2d}}{v}\right)\cdot\left(\frac{b,2}{2}\right).$$

By Hilbert reciprocity law, we have

$$\left(\frac{a-\sqrt{2d}, x_0-y_0\sqrt{2d}}{v}\right)=\prod_{\mathfrak{p}\neq v}\left(\frac{a-\sqrt{2d}, x_0-y_0\sqrt{2d}}{\mathfrak{p}}\right).$$

Since

$$(a-\sqrt{2d})(a+\sqrt{2d})=-2b^2 \quad \text{and} \quad (x_0-y_0\sqrt{2d})(x_0+y_0\sqrt{2d})=2z_0^2$$

with $(a,b)=1$ and $(x_0,y_0,z_0)=1$ respectively, one has

$$\mathrm{ord}_{\mathfrak{p}}(a-\sqrt{2d})\equiv \mathrm{ord}_{\mathfrak{p}}(x_0-y_0\sqrt{2d})\equiv 0 \bmod 2$$

for $\mathfrak{p}\neq v$ and $\mathfrak{p}<\infty_E$. Since $x_0-y_0\sqrt{2d}>0$ over $\mathfrak{p}\in\infty_E$, one obtains that

$$\left(\frac{a-\sqrt{2d}, x_0-y_0\sqrt{2d}}{\mathfrak{p}}\right)=1$$

for all $\mathfrak{p}\neq v$. So

$$\left(\frac{\eta, x_0-y_0\sqrt{2d}}{v}\right)=\left(\frac{b,2}{2}\right).$$

Since $a^2+2b^2=2d$, we have

$$\left(\frac{d}{l}\right)=1 \text{ for any odd prime } l|a$$

and

$$\left(\frac{2d}{l}\right)=1 \text{ for any odd prime } l|b.$$

Therefore

$$\prod_{p|d}\left(\frac{a}{p}\right)=\prod_{p|d}\left(\frac{a,d}{p}\right)=\left(\frac{a,d}{2}\right)\prod_{\text{odd } l|a}\left(\frac{a,d}{l}\right)=1\cdot 1=1,$$

since $d\equiv 1 \bmod 8$, the second equation holds by Hilbert reciprocity law. And

$$\begin{aligned}\left(\frac{b,2}{2}\right)&=\left(\frac{b,2d}{2}\right)=\prod_{l|b}\left(\frac{b,2d}{l}\right)\cdot\prod_{p|d}\left(\frac{b,2d}{p}\right)\\&=1\cdot\prod_{p|d}\left(\frac{b}{p}\right)=\prod_{p|d}\left(\frac{b/a}{p}\right).\end{aligned}$$

Since $a^2+2b^2=2d$, we have $(a/b)^2\equiv -2 \bmod p$ for any $p|d$. Hence

$$\left(\frac{a/b}{p}\right)=\left(\frac{-2}{p}\right)_4 \text{ for any } p|d.$$

Since $p\equiv 1 \bmod 8$, then $(\frac{-2}{p})_4=(\frac{2}{p})_4$. So we have

$$\left(\frac{\eta, x_0-y_0\sqrt{2d}}{v}\right)=\left(\frac{b,2}{2}\right)=\prod_{p|d}\left(\frac{b/a}{p}\right)=\prod_{p|d}\left(\frac{2}{p}\right)_4=\left(\frac{2}{d}\right)_4.$$ □

Proposition 4.4. *Suppose that $p_i\equiv 1 \bmod 8, 1\leqslant i\leqslant m$ are distinct primes. Let $d=p_1p_2\cdots p_m$. Then*

(1) *If there exist two integers $r,s\equiv\pm 3 \bmod 8$ such that $2d=r^2+s^2$, then the equation $x^2-2dy^2=-1$ is not solvable over $\mathbb{Z}$.*

(2) *If the equation $x^2-2dy^2=-2$ is solvable over $\mathbb{Z}$, then $\left(\frac{2}{d}\right)_4=1$.*

Proof. (1) By Lemma 3.2, one has $(\frac{\xi, x_0-y_0\sqrt{2d}}{v}) = -1$ for any $\xi \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi) = -1$. This implies that

$$\psi_{\Theta/E}\Big(f_E\Big[\prod_{p\leqslant\infty}(x_p, y_p)\Big]\Big) = \left(\frac{x_2+y_2\sqrt{2d}, x_0-y_0\sqrt{2d}}{v}\right) = -1$$

for any $\prod_{p\leqslant\infty}(x_p, y_p) \in \prod_{p\leqslant\infty}\mathbf{X}_{(-1)}(\mathbb{Z}_p)$. Then $x^2-2dy^2=-1$ is not solvable over $\mathbb{Z}$ by Theorem 1.3.

(2) With similar argument as above, the result follows from Lemma 4.3. □

Remark. For the case (1), when d is the product of two distinct primes, the unsolvability of $x^2-2dy^2=-1$ was proved by Epstein [5] and Rédei [12].

Let $d=p$ be a prime. If $p \not\equiv 1 \bmod 8$, the solvability of the three equations

$$x^2-2py^2=-1, \quad x^2-2py^2=\pm 2$$

is well known (see [4] or [14]). If $p \equiv 1 \mod 8$, the solvability problem is more complicated. The following result gives a new proof of Theorems 3 and 4 in [10].

Corollary 4.5. *Let p be an odd prime.*

(1) *Let $p \equiv 9 \mod 16$. If $(\frac{2}{p})_4 = -1$, then $x^2-2py^2=-1$ is solvable over $\mathbb{Z}$. If $(\frac{2}{p})_4 = 1$, then $x^2-2py^2=-2$ is solvable over $\mathbb{Z}$.*

(2) *Let $p \equiv 1 \mod 16$. If $(\frac{2}{p})_4 = -1$, then $x^2-2py^2=2$ is solvable over $\mathbb{Z}$.*

Proof. Since p is an odd prime, one and only one of the three equations

$$x^2-2py^2=-1, \quad x^2-2py^2=\pm 2$$

is solvable over $\mathbb{Z}$ (see [4, p. 225]).

(1) Since $p \equiv 9 \mod 16$, one has the equation $x^2-2py^2=2$ is not solvable over $\mathbb{Z}$ by Corollary 4.2.

If $(\frac{2}{p})_4 = -1$, then $x^2-2py^2=-2$ is not solvable over $\mathbb{Z}$ by Proposition 4.4. Therefore $x^2-2py^2=-1$ is solvable.

Suppose $(\frac{2}{p})_4 = 1$. Let v be the unique place of E over 2. Let $\xi_1, \xi_2 \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi_1) = -2$ and $N_{E_v/\mathbb{Q}_2}(\xi_2) = 2$. By Lemma 4.1 and 4.3, we have

$$\left(\frac{\xi_1, x_0-y_0\sqrt{2p}}{v}\right) = \left(\frac{2}{p}\right)_4 = 1 \text{ and } \left(\frac{\xi_2, x_0-y_0\sqrt{2p}}{v}\right) = -1.$$

By Lemma 3.1, we have

$$\left(\frac{\xi, x_0-y_0\sqrt{2p}}{v}\right) = \left(\frac{\xi_1/\xi_2, x_0-y_0\sqrt{2p}}{v}\right) = 1\cdot(-1) = -1$$

for any $\xi \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi) = -1$, since $N_{E_v/\mathbb{Q}_2}(\xi_1/\xi_2) = -1$. With the similar arguments as in the proof of Proposition 4.4, one has $x^2-2py^2=-1$ is not solvable over $\mathbb{Z}$. Therefore $x^2-2py^2=-2$ is solvable over $\mathbb{Z}$.

(2) Suppose $p \equiv 1 \mod 16$. Since $(\frac{2}{p})_4 = -1$, one has that $x^2-2py^2=-2$ is not solvable over $\mathbb{Z}$ by Proposition 4.4. Let v be the unique place of E over 2. Let $\xi_1, \xi_2 \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi_1) = -2$ and $N_{E_v/\mathbb{Q}_2}(\xi_2) = 2$. By Lemmas 4.1 and 4.3, we have

$$\left(\frac{\xi_1, x_0-y_0\sqrt{2p}}{v}\right) = \left(\frac{2}{p}\right)_4 = -1 \text{ and } \left(\frac{\xi_2, x_0-y_0\sqrt{2p}}{v}\right) = 1.$$

By Lemma 3.1, we have

$$\left(\frac{\xi, x_0-y_0\sqrt{2p}}{v}\right) = \left(\frac{\xi_1/\xi_2, x_0-y_0\sqrt{2p}}{v}\right) = (-1)\cdot 1 = -1$$

for any $\xi \in E_v^*$ with $N_{E_v/\mathbb{Q}_2}(\xi) = -1$. Therefore $x^2-2py^2=-1$ is not solvable over $\mathbb{Z}$. Then $x^2-2py^2=2$ is solvable over $\mathbb{Z}$. □

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